INCH-POUND

MIL-A-52100C
7 October 1992
SUPERSEDING
MIL-A-52100B
21 September 1971

MILITARY SPECIFICATION

ADAPTERS: FOR BRIDGE, FLOATING, ALUMINUM, HIGHWAY-TYPE, DECK-BALK SUPERSTRUCTURE ON PNEUMATIC FLOATS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers adapters for use with the bridge, floating, aluminum, highway-type, deck-balk superstructure.

1.2 Classification. The adapters shall be of the following types, as specified:

Type I - Adapter, saddle, deck balk, normal
Type II - Adapter, saddle, deck balk, offset
Type III - Adapter, curb, deck balk

2. APPLICABLE DOCUMENTS

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	- Colors used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other drawings. The following drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13218E9941	- Adapter, Saddle, Deck Balk, Steel, Normal.
TA13218E9942	- Adapter, Saddle, Deck Balk, Steel, Offset.
TA13208E9900	- Adapter, Curb, Deck Balk.
13217E7895	- Adapter, Saddle, Deck Balk, Steel, Normal, Packaging of.
13217E7896	- Adapter, Saddle, Deck Balk, Steel, Offset, Packaging of.
13217E7897	- Adapter, Curb, Deck Balk, Packaging of.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-JB, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

AMERICAN WELDING SOCIETY (AWS)

A 5.1	- Covered Carbon Steel Arc-Welding Electrodes.
A 5.5	- Low-Alloy Steel Covered Arc-Welding Electrodes.
D 1.1	- Structural Welding Code-Steel.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualifications Standard For Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 2200 Mill Road, Alexandria, VA 22314.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The adapters shall be in accordance with the drawings specified in 3.8 through 3.10, as applicable, and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operating of the adapter. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings.

3.3.1 Material deterioration prevention and control. The adapter shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the adapter may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Welding electrodes. Welding electrodes for structural steel shall conform to AWS A 5.1, E60XX series. Welding electrodes for low-alloy steel shall conform to AWS A 5.5, E70XX series.

3.5 Identification marking. The adapters shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Cleaning, treatment, and painting. All parts of the adapters shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that the color number designation shall be 34088 in accordance with FED-STD-595. Contact surfaces, regardless of whether such contact is between similar or dissimilar metals, shall be cleaned, treated, and painted prior to assembly.

3.7 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each type I and type II adapter. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.8 Type I adapter, normal. The type I adapter shall be as shown on drawing TA13218E9941 and as specified herein.

3.9 Type II adapter, offset. The type II adapter shall be as shown on drawing TA13218E9942 and as specified herein.

3.10 Type III adapter, curb. The type III adapter shall be as shown on drawing TA13208E9900 and as specified herein.

3.11 Workmanship.

3.11.1 Steel fabrication. Steel used in the fabrication of the adapter shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs, rough edges, and edges of holes shall be ground smooth.

3.11.2 Welders and welding.

3.11.2.1 Welders and welding operator. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.1 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.5).

3.11.2.2 Welding. The surfaces of parts to be welded shall be in accordance with ASME or AWS code. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the D 1.1. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.11.2.3 Jigs and fixtures. Components, except for minor parts, shall be assembled in steel jigs or frames and welded while securely held in position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

Minor parts may be held securely and tightly in position by any means that will insure secure and proper positioning. The jigs or frames shall be designed so that distortion of the parts will be held at a minimum.

3.11.3 Interchangeability. Like adapters shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance or rejection (see 6.4 and 6.6).

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Examination. The first article adapter shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The adapters shall be examined for the following defects:

Major

101. Material not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
105. Workmanship not as specified (see 3.11).
106. Incomplete or incorrect assembly (see 3.1).
107. Any part or component not in accordance with QAP requirement as shown on the drawings (see 4.1.2).
108. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal (see 3.11.2.2).
109. Welders and welding operators not certified as specified (see 3.11.2.1).
110. Welding and welds not as specified (see 3.11.2).
111. Interchangeability not as specified (see 3.11.3).
112. Jigs and fixtures not as specified (see 3.11.2.3).

Minor

201. Dimensions other than those identified by QAP data not as specified (see 4.1.3).
202. Identification marking incorrect, missing, or illegible (see 3.5).
203. Cleaning, treatment, and painting not as specified (see 3.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

204. Stenciling not as specified for type I and type II adapters (see 3.7).

4.6 Inspection of packaging.

4.6.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.3 Examination. Samples selected in accordance with 4.6.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

113. Type I adapters not nested and bundled together in the quantities required and in accordance with the referenced drawings (see 5.1.1.1).
114. Type II adapters not nested and bundled together in the quantities required and in accordance with the referenced drawings (see 5.1.1.2).
115. Type III adapters not packed in the quantities required and in accordance with the referenced drawings for level A or B (see 5.1.1.3).
116. Marking illegible, incorrect, or incomplete (see 5.2).
117. Packing not as specified (see 5.1).

5. PACKAGING

5.1 Packing. Packing shall be level A, B or C as specified.

5.1.1 Level A.

5.1.1.1 Type I adapters. Five type I adapters shall be nested and bundled as shown in drawing 13217E7895.

5.1.1.2 Type II adapters. Four type II adapters shall be nested and bundled as shown in drawing 13217E7896.

5.1.1.3 Type III adapters. Eight type III adapters shall be packed in a close-fitting box as specified for level A as shown in drawing 13217E7897.

5.1.2 Level B.

5.1.2.1 Type I and type II adapters. The type I and type II adapters shall be packed as specified for level A.

5.1.2.2 Type III adapters. Eight type III adapters shall be packed in a close-fitting box as specified for level B as shown in drawing 13217E7897.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

5.1.3 Level C.

5.1.3.1 Type I and type II adapters. The type I and type II adapters shall be packed as specified for level A.

5.1.3.2 Type III adapters. The type III adapters shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in a wooden container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Shipping containers and bundles shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The saddle adapters, both normal and offset, are to be used as intermediate supports between aluminum deck-balk superstructures and 24-ton pneumatic floats. The curb adapter is to be used to raise a row of balk to provide curbing approximately 6 inches above the normal deck elevation. This item is a component of the following supply catalogs: SC 5420-97CL-E35, E36 and E41.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type of adapter required (see 1.2).
d. Time frame required for submission of first article (see 3.2).
e. Level of packing required (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction item (see 3.2), and the number of items to be tested specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

6.4 Quality assurance provisions (QAP). The contracting officer should require the supplier to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP Characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrected action taken when applicable."

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.11.2.1	DI-MISC-80876	Welding Procedure Qualification Test Report.
3.11.21	DI-MISC-80875	Welding Procedures

6.6 Definitions.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements Section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 Subject term key word listing.

Bridge Adapter
Bridge Connections
Gangplank

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 5420-0197

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-52100C	2. DOCUMENT DATE (YYMMDD) 921007

3. DOCUMENT TITLE
Adapters: For Bridge, Floating, Aluminum, Highway-Type, Deck-Balk Superstructure on Pneumatic Floats

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Geraldine F. Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3469	(2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.